

Exercise #1 on XML

Christian Rinderknecht

9 October 2006

1 Threesome Shuffle

This time it is about shuffling three stacks of cards in the following way (the top of the stacks is on the top of the figure):

$$\begin{array}{ccccc} a_1 & a_2 & a_3 & & a_3 \\ b_1 & b_2 & b_3 & \rightarrow \dots \rightarrow & a_1 \\ c_1 & c_2 & & & a_2 \\ & d_2 & & & b_3 \\ & & & & b_1 \\ & & & & b_2 \\ & & & & c_1 \\ & & & & c_2 \\ & & & & d_2 \end{array}$$

If the three stacks are empty, then the result is empty too.

The empty stack is noted as usual `EMPTY` and the stack S with item e on the top is noted `PUSH( $e$ ,  $S$ )`.

Question. Define $\text{TSS}(S_1, S_2, S_3)$ as rewriting to the shuffling.

Question. You will get extra points if

- your function is efficient;
- you make another version which returns `ERROR` if the number of items in the three stacks is not the same.